Voice-activated home automation using Arduino Mega & HC-05 Bluetooth. MSc Thesis NTU 2024.

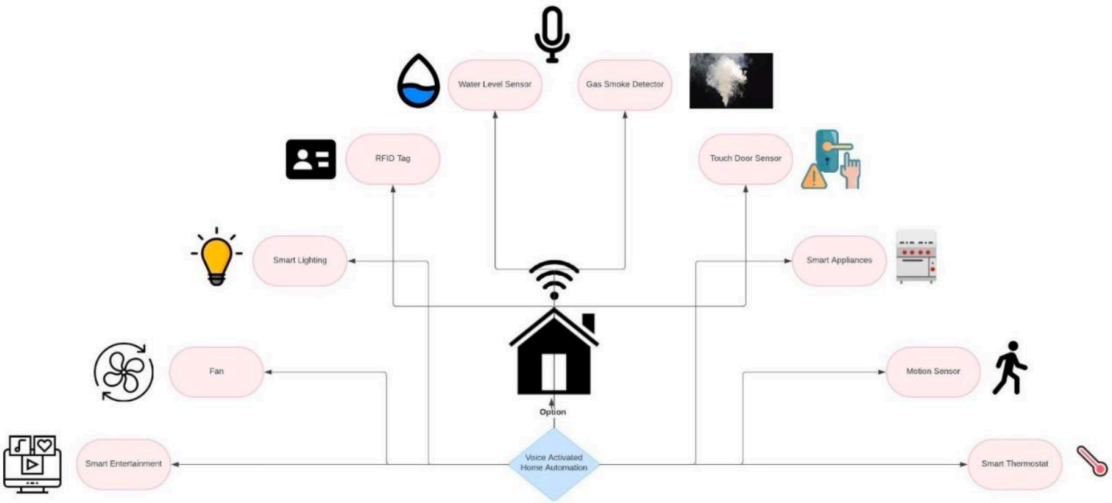


Figure 1: Home automation

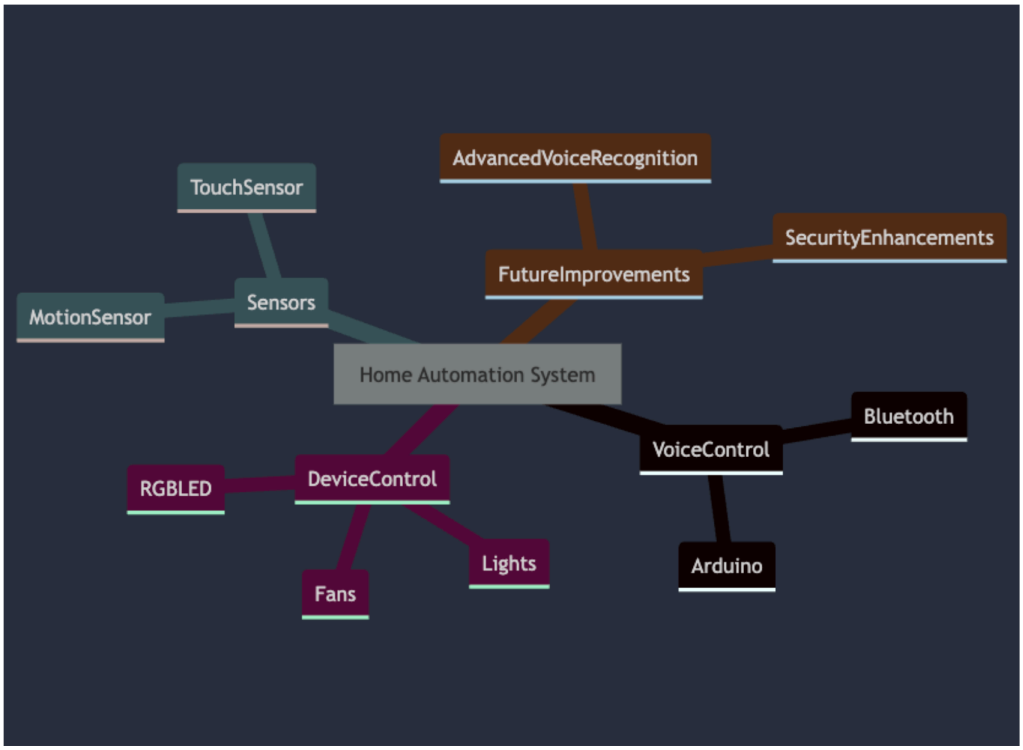


Figure:2 – Mind Map of Voice-activated Home Automation

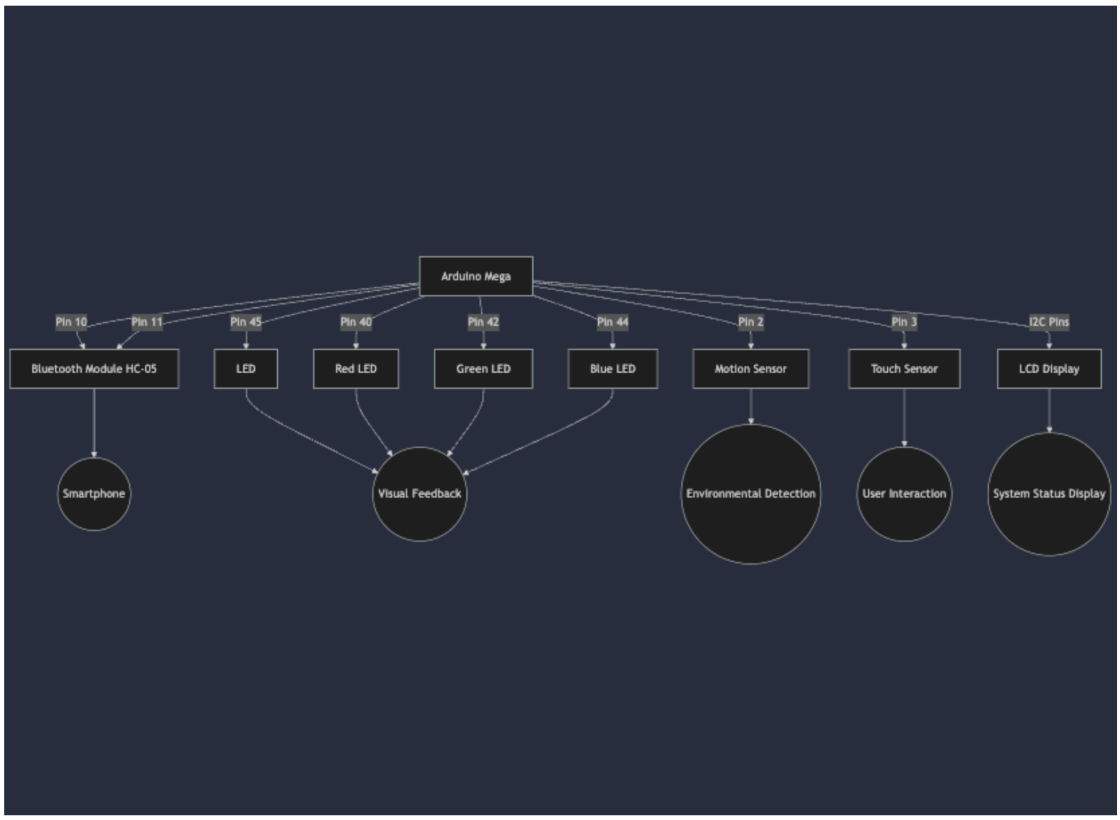


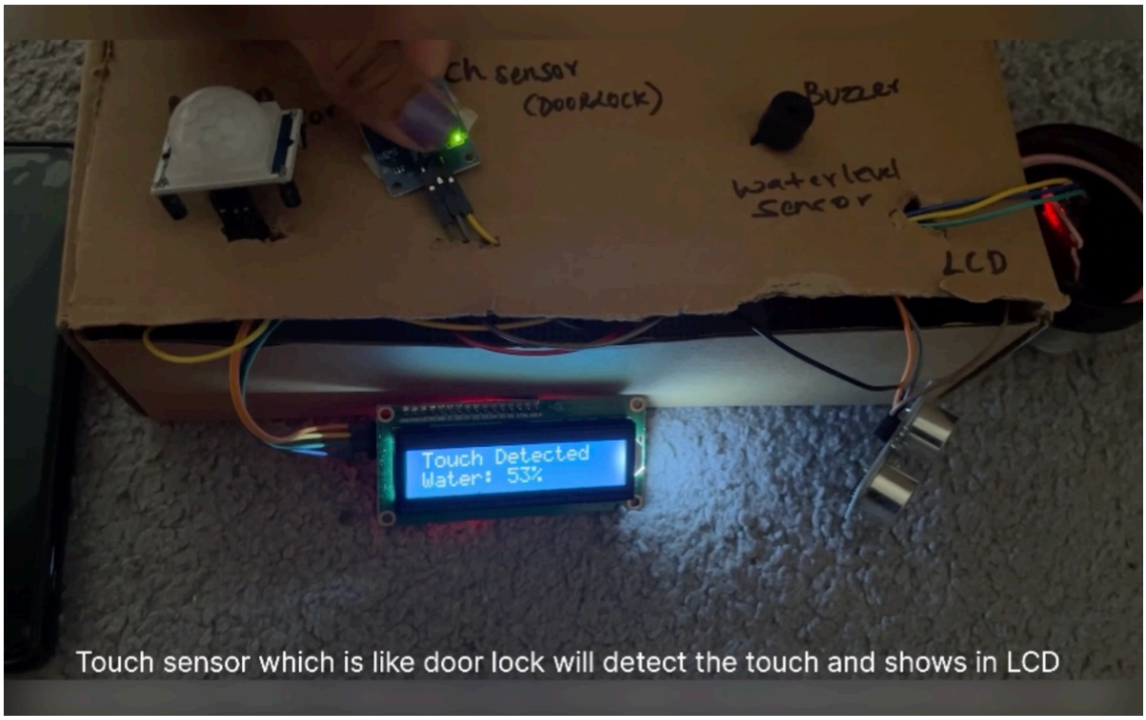
Figure 3: Overview of hardware working

Table: 1 Test plan

Test Scenario	Expected Outcome	Actual Outcome	Status
Voice command: "Turn on light"	Light turns on	Light turned on	Pass
Voice command: "Turn off fan"	Fan turns off	Fan turned off	Pass
Sensor trigger: Motion detected	Light turns on automatically	Light turned on	Pass
Background noise interference test	Voice command recognized accurately	Misinterpreted command, no action	Fail (Noise)

Table: 2 System accuracy

Survey Question	Average Rating (1-5)
Ease of use for controlling devices	4.7
Responsiveness of the system	4.3
Clarity of feedback on LCD display	4.5
Overall satisfaction with the system	4.6



LCD Output - For Touch detection